BLOCKCHAIN TECHNOLOGY

Survey Report on Types of Blockchains and Their Real-Time Use Cases

Blockchain Technology Practical | Report + Rough Work + Viva Notes

Practical No.	1
Aim	To study the types of blockchain and their real-time applications.
Subject	Blockchain Technology
Student Name	_____
Class / Division	_____
Roll No.	_____
Date	_____

Submitted for academic practical and viva preparation

1. Aim

To prepare a survey report on the major types of blockchain technology, understand their characteristics and architecture, and study their real-time use cases in different industries.

2. Objectives

• To understand the basic concept and working principles of blockchain.

• To identify and compare public, private, consortium, and hybrid blockchains.

• To study the benefits, limitations, and suitability of each type.

• To explore real-world applications in finance, healthcare, supply chains, education, and government.

• To understand how blockchain and smart contracts support trusted digital systems.

3. Abstract

Blockchain is a distributed ledger technology that records transactions in a sequence of cryptographically linked blocks. Instead of relying only on a central database, blockchain allows multiple participants to maintain and validate a shared record according to agreed network rules. This survey examines four widely discussed blockchain types—public, private, consortium, and hybrid—and compares them based on access, governance, transparency, privacy, and performance. It also surveys practical applications such as digital payments, supply-chain traceability, healthcare information exchange, academic credential verification, and smart contracts. The report concludes that blockchain is not a single universal solution; the appropriate architecture depends on the application's trust model, privacy requirements, participants, and performance needs.

4. Introduction

Traditional databases are commonly managed by a central organization that controls access and updates. This arrangement works well in many applications, but it can create challenges when multiple independent organizations need to share records and trust the same information. Blockchain provides an alternative approach: participants maintain a shared ledger, and updates are accepted through a protocol or governance process.

A blockchain consists of blocks containing data and metadata. Each block typically includes a cryptographic hash of the previous block, creating a linked structure. If information in an earlier block is changed, its hash no longer matches the value referenced by later blocks. This makes unauthorized changes detectable and gives blockchain its tamper-evident property.

Blockchain does not automatically guarantee that every input is true, that data is private, or that a system cannot be attacked. Its security depends on cryptography, consensus or validation rules, software quality, participant behavior, and governance. Therefore, blockchain should be selected only when its distributed trust and auditability features are useful for the problem.

5. Basic Architecture and Working of Blockchain

The basic elements of a blockchain system are:

• **Transaction:** A record of an action, such as transferring an asset or updating a status.

• **Block:** A container that groups verified transactions with metadata.

• **Hash:** A fixed-length cryptographic digest used to identify data and detect changes.

• **Previous-block hash:** A reference connecting one block to the preceding block.

• **Nodes:** Computers or participants that store, verify, or communicate blockchain data.

• **Consensus / validation mechanism:** Rules used to agree on which transactions or blocks are accepted.

• **Ledger:** The ordered record of accepted blocks and transactions.

Typical transaction flow

1. A user initiates a transaction and digitally signs it where required.

2. The transaction is broadcast to participating nodes.

3. Nodes check whether it follows the network rules, including signature and balance or permission checks.

4. Valid transactions are grouped into a block or recorded through the network's ordering process.

5. The network's consensus or validation process accepts the block or update.

6. Nodes update their ledger copies, and the transaction becomes part of the recorded history.

> The exact process differs between blockchain platforms. For example, public networks may use Proof of Work or Proof of Stake, while permissioned networks may use voting, endorsement, or other enterprise-oriented protocols.

6. Types of Blockchains

Blockchain networks are commonly classified according to who can read the ledger, submit transactions, validate blocks, and govern the network. The four major categories are public, private, consortium, and hybrid.

6.1 Public Blockchain

Definition: A public blockchain is generally an open, permissionless network. Anyone can usually read public records and may participate in transaction submission or validation according to the network protocol.

Characteristics:

• Open participation and broad accessibility.

• High transparency for on-chain records, although user identities may be represented by addresses.

• No single organization normally controls the entire network.

• Can use consensus mechanisms such as Proof of Work (PoW) or Proof of Stake (PoS).

• Public networks may face throughput, latency, transaction-cost, and privacy challenges.

Examples: Bitcoin and Ethereum are prominent public blockchain networks.

Real-time use cases: Cryptocurrency transfers; public settlement and tokenization; decentralized finance (DeFi); non-fungible tokens (NFTs); decentralized applications; and publicly verifiable records.

Example scenario: A person transfers a digital asset to another person. The transaction is broadcast to the network, checked under the protocol rules, and included in the public ledger. Participants can independently verify the transaction history.

Advantages: Open participation, strong transparency, and reduced dependence on a single operator. **Limitations:** Public visibility may conflict with privacy needs, and performance or fees may vary with network demand.

6.2 Private Blockchain

Definition: A private blockchain is a permissioned network controlled by one organization or a clearly defined operator. Participants must receive authorization to access particular functions.

Characteristics:

• Access is restricted and roles can be assigned to users.

• The organization can control membership and permissions.

• Transaction visibility can be limited to authorized participants.

• It may provide higher throughput and lower latency than some public networks, depending on its design.

• Governance and trust are more centralized than in a public blockchain.

Examples: Private enterprise networks built with Hyperledger Fabric or MultiChain. Hyperledger Fabric is a platform for permissioned blockchain networks and can be configured for different enterprise requirements.

Real-time use cases: Internal audit trails, enterprise asset tracking, interdepartmental record sharing, controlled supply-chain systems, and business process management.

Example scenario: A company maintains a shared ledger for tracking inventory between its warehouses. Only approved employees and systems can submit updates or view sensitive fields.

Advantages: Access control, privacy options, and organizational governance. **Limitations:** A single organization has significant control, so the network may not provide the same decentralization or independent trust as a public blockchain.

6.3 Consortium Blockchain

Definition: A consortium blockchain, also called a federated blockchain in some contexts, is governed by a group of organizations. Selected members may operate nodes or participate in transaction validation.

Characteristics:

• Permissioned participation with multiple organizational members.

• Governance and validation responsibilities are shared among consortium participants.

• Sensitive data can be restricted to relevant parties.

• Useful when several organizations need a common record but do not want one member to control it entirely.

• Requires agreements about membership, data access, dispute resolution, and network rules.

Examples: Consortium networks can be implemented using Hyperledger Fabric. R3 Corda is used in financial and enterprise networks and supports permissioned, multi-party workflows; specific deployments differ in architecture.

Real-time use cases: Interbank settlement, trade finance, shared logistics records, insurance coordination, and healthcare data exchange between institutions.

Example scenario: Several banks share a permissioned network to reconcile selected transaction records. Each bank retains its own systems while the consortium provides a common, auditable view of agreed records.

Advantages: Shared governance, controlled access, and coordination between organizations. **Limitations:** Setup requires collaboration, governance agreements, and trust among participating

members.

6.4 Hybrid Blockchain

Definition: A hybrid blockchain combines public and private or permissioned elements. An organization can keep selected records private while exposing limited information or proofs for public verification.

Characteristics:

• Combines controlled access with selective transparency.

• Sensitive data can remain in a restricted environment.

• Public verification may be supported through hashes, proofs, or selected on-chain information.

• The design can be tailored to business, legal, and privacy requirements.

• The term 'hybrid' can describe different architectures, so implementations should be examined individually.

Examples: Hybrid architectures can be built by combining private systems with public blockchain anchoring. XinFin / XDC Network is often discussed in hybrid public-private blockchain contexts; deployments and features should be checked for the particular use case.

Real-time use cases: Land and property records, business registries, healthcare systems, identity or credential verification, and supply chains that need private operational data with public auditability.

Example scenario: A hospital stores patient details in a restricted system but publishes a cryptographic proof or hash to a public blockchain. The public record can help verify that a record existed and was not altered without revealing the medical details.

Advantages: Flexible privacy and selective transparency. **Limitations:** Architecture can be more complex, and integration between private and public components must be carefully designed.

7. Comparative Analysis of Blockchain Types

Feature	Public	Private	Consortium	Hybrid
Access	Open	Authorized users	Selected members	Mixed
Governance	Network-wide	One organization	Several organizations	Combined
Transparency	Generally high	Restricted	Controlled	Selective
Decentralization	Usually highest	Lower	Moderate	Varies
Privacy	Limited by default	Configurable	Configurable	Selective
Performance	Varies by network	Often optimized	Often optimized	Depends on design
Best suited for	Open digital assets	Single enterprise	Cross-organization work	Privacy + verification

No blockchain type is universally best. Public networks prioritize openness and broad verification; private networks prioritize organizational control; consortium networks support shared governance; and hybrid designs aim to balance privacy with verifiability.

8. Real-Time Use Cases of Blockchain

8.1 Banking and Financial Services

Blockchain can support digital asset transfers, cross-border payment experiments, settlement, tokenization, and shared transaction records. Distributed records can help participants reconcile transactions and improve auditability. Permissioned networks are often considered where financial institutions need controlled membership and confidentiality. However, adoption depends on regulations, interoperability, settlement finality, and integration with existing financial systems.

8.2 Supply Chain and Logistics

Supply chains involve manufacturers, distributors, transporters, retailers, and customers. Blockchain can record events such as product creation, shipment, inspection, and delivery. A shared traceability record can help identify the source of goods, reduce disputes, and support product recalls. For example, food or pharmaceutical supply chains can record custody and verification events. Blockchain cannot guarantee that a physical item or initial data entry is genuine; reliable sensors, inspections, and trusted participants remain important.

8.3 Healthcare

Healthcare organizations may use blockchain to coordinate access, audit data sharing, verify integrity, or manage consent-related events. Patient information should generally remain in secure, compliant storage rather than being placed directly on a public blockchain. Blockchain may store references, permissions, or integrity proofs while sensitive records remain off-chain. Privacy, consent, interoperability, and legal requirements are essential.

8.4 Education and Digital Certificates

Universities and certification authorities can issue digital credentials that are verifiable by employers or other institutions. Blockchain can provide a tamper-evident record or verification mechanism, reducing the risk of forged certificates and simplifying credential checks. The credential itself may be stored digitally with a verification proof recorded on-chain.

8.5 Government and Public Records

Blockchain can support audit trails for selected public services, land-record verification, public procurement tracking, and digital identity-related systems. A shared ledger may improve transparency and reduce unauthorized record changes. These systems still require legal authority, identity verification, privacy protection, and mechanisms for correcting errors.

8.6 Smart Contracts and Decentralized Applications

A smart contract is software deployed on a blockchain that executes predefined rules when called and when the required conditions are met. Smart contracts can automate escrow, token transfers, decentralized finance operations, digital asset management, and selected insurance workflows. They execute code as written, so bugs or incorrect assumptions can cause unintended outcomes. Auditing and testing are therefore important.

8.7 Real Estate and Asset Tokenization

Blockchain may be used to represent ownership claims or rights to assets through digital tokens, maintain transaction histories, and support record coordination. Tokenization does not by itself transfer legal ownership; the token's legal meaning depends on applicable laws, contracts, and the system's governance.

9. Advantages of Blockchain

• **Distributed recordkeeping:** Multiple participants can maintain a shared ledger.

• **Tamper evidence:** Cryptographic linking helps detect unauthorized changes.

• **Transparency and auditability:** Authorized participants can trace recorded activity.

• **Traceability:** Transactions and asset histories can be followed across stages.

• **Reduced reconciliation:** A shared source of truth may reduce mismatched records between organizations.

• **Programmability:** Smart contracts can automate certain rules and workflows.

10. Limitations and Challenges

• **Scalability:** Some networks cannot process as many transactions as centralized systems.

• **Energy consumption:** Proof of Work can consume substantial energy; other consensus designs have different resource profiles.

• **Privacy:** Public ledgers may expose transaction details or metadata.

• **Regulation:** Legal requirements vary by jurisdiction and application.

• **Data quality:** Blockchain preserves submitted records but cannot guarantee that external data is correct.

• **Immutability and correction:** Errors may require additional corrective transactions or governance procedures.

• **Security risks:** Smart contract bugs, stolen keys, software vulnerabilities, and governance attacks remain possible.

• **Cost and complexity:** Development, integration, operation, and governance can be expensive.

11. Survey Findings and Discussion

The survey shows that blockchain architecture should be selected according to the trust relationships among participants. Public blockchains are appropriate when open participation and public verification are important. Private blockchains suit organizations that need controlled access and centralized administration. Consortium blockchains are useful when several independent organizations need a shared system with joint governance. Hybrid designs are suitable when private information must be protected while selected records or proofs remain publicly verifiable.

A further finding is that blockchain is most valuable when multiple parties need a shared, auditable record and do not want to rely entirely on one operator. If a conventional database already meets the requirements efficiently and the organization is trusted by all participants, blockchain may add unnecessary complexity. The decision should consider performance, privacy, compliance, cost, and governance—not only the technology's popularity.

12. Conclusion

Blockchain is a distributed ledger technology that can improve shared recordkeeping, traceability, and tamper evidence. This report surveyed public, private, consortium, and hybrid blockchain types and their characteristics. It also examined applications in finance, supply chains, healthcare, education, government, smart contracts, and real estate. Each type offers a different balance of openness, control, privacy, and governance. The most suitable blockchain depends on the application's requirements and the relationship between its participants. Blockchain is not a replacement for every database, but it can be useful where shared trust and verifiable records are important.

13. References / Further Reading

• Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System.*

• Ethereum.org. *Ethereum documentation* — https://ethereum.org/en/developers/docs/

• Hyperledger Fabric Documentation — https://hyperledger-fabric.readthedocs.io/

• NIST. *Blockchain Technology Overview*, NISTIR 8202 — https://doi.org/10.6028/NIST.IR.8202

• IBM. *What is blockchain?* — https://www.ibm.com/topics/blockchain

• World Health Organization and relevant national regulations should be consulted for healthcare-specific data protection requirements.

ROUGH WORK

Topic: Survey Report on Types of Blockchains and Real-Time Use Cases

Blockchain: Distributed ledger; records transactions in blocks; blocks linked using cryptographic hashes; shared among network participants.

Types — remember:

Type	Control / access	Example / use
Public	Open participation	Bitcoin, Ethereum; crypto, DeFi
Private	One organization	Enterprise records, internal supply chain
Consortium	Multiple selected organizations	Banks, trade finance, shared logistics
Hybrid	Private + selected public verification	Land records, credentials, healthcare proofs

Working: Identify blockchain types → compare access, governance, privacy, and transparency → map each type to suitable real-world applications → note advantages and limitations → conclude that selection depends on requirements.

Applications: Finance • Supply chain • Healthcare • Education • Government • Smart contracts • Real estate

Advantages: Shared ledger • Transparency • Traceability • Tamper evidence • Auditability

Limitations: Scalability • Privacy • Energy (PoW) • Regulation • Cost • Incorrect input data

One-line conclusion: Blockchain improves trusted and auditable recordkeeping, but the right type depends on access, privacy, governance, and performance requirements.

VIVA EXPLANATION (MEDIUM-SIZED)

This practical was a survey, so we did not need to build a blockchain. We studied what blockchain is, how blocks are linked using cryptographic hashes, and why a distributed ledger can help different participants share records. We then compared four types: public, private, consortium, and hybrid. Public blockchain is open to participants; private blockchain is controlled by one organization; consortium blockchain is governed by several organizations; and hybrid blockchain combines private data handling with selected public verification. We studied their applications in banking, supply chains, healthcare, education, government, and smart contracts. The main conclusion is that blockchain is useful when parties need a shared and tamper-evident record, but the type should be chosen based on privacy, access, governance, performance, and cost.

VIVA QUESTIONS AND ANSWERS

1. What is blockchain?
Blockchain is a distributed ledger that stores records in cryptographically linked blocks and is maintained by network participants.

2. Why is it called a blockchain?
Because data is grouped into blocks, and each block is linked to the previous block using its hash.

3. What is a distributed ledger?
A ledger whose records are shared and maintained across multiple participants or nodes.

4. What is a hash?
A fixed-length cryptographic output generated from data. A change in the input normally produces a different hash.

5. What is the role of the previous-block hash?
It links the current block to the previous block and helps make changes to earlier records detectable.

6. What are the four main blockchain types?
Public, private, consortium, and hybrid.

7. What is a public blockchain?
An open blockchain where participation is generally permissionless. Bitcoin and Ethereum are examples.

8. What is a private blockchain?
A permissioned blockchain controlled by one organization, with access limited to authorized users.

9. What is a consortium blockchain?
A blockchain governed by a group of selected organizations rather than a single organization.

10. What is a hybrid blockchain?
A design combining private or permissioned components with selected public visibility or verification.

11. Which blockchain is most decentralized?
Public blockchains are generally the most decentralized among these categories, although decentralization varies by implementation.

12. Which blockchain is suitable for a group of banks?
A consortium or permissioned blockchain can be suitable because banks can share governance and restrict access.

13. What is consensus?
A process or protocol that allows network participants to agree on accepted transactions or blocks.

14. What is Proof of Work?
A consensus mechanism in which participants compete to solve a computational puzzle to propose blocks, as used by Bitcoin.

15. What is Proof of Stake?
A consensus mechanism where validators are selected and incentivized based partly on stake and protocol rules.

16. What is a smart contract?
A program deployed on a blockchain that executes predefined instructions when called and conditions are satisfied.

17. Give two real-time blockchain use cases.
Cryptocurrency payments and supply-chain traceability. Other examples include digital certificates and smart contracts.

18. How is blockchain used in healthcare?
It can support audit trails, controlled data sharing, and integrity verification. Sensitive patient data should be protected and generally kept off public chains.

19. How can blockchain help education?
It can support verifiable digital certificates and reduce the risk of forged academic credentials.

20. What is the main advantage of blockchain?
A shared, tamper-evident, and auditable record that reduces dependence on a single central record keeper.

21. What are the limitations of blockchain?
Scalability, privacy, regulation, cost, energy use in some systems, and the possibility of incorrect data being entered.

22. Is blockchain completely secure?
No. It offers security features, but vulnerabilities in software, smart contracts, keys, and governance can still be exploited.

23. Is blockchain always better than a database?
No. A conventional database may be better when one trusted organization controls the data and high performance or simple management is required.

24. What is immutability?
The property that recorded data is difficult to alter without detection. It does not mean errors can never be corrected.

25. What did you conclude from this practical?
Different blockchain types suit different needs. The choice depends on access, privacy, governance, transparency, and performance.

LAST-MINUTE REVISION

Public = open | **Private = one organization** | **Consortium = multiple organizations** | **Hybrid = private + public features**

For viva, explain the definition first, then the four types, one example/use case, and one limitation. Keep the answer clear and avoid claiming that blockchain is perfectly secure or that every real-world record is automatically trustworthy.